

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2456948.6689 +/- 0.0026 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.135 +/- 0.012
Transit depth (Rp/Rs)^2	1.83 +/- 0.32 [%]
Orbital Inclination (inc)	85.25 +/- 0.99
Airmass coefficient 1 (a1)	1.0345 +/- 0.0078
Airmass coefficient 2 (a2)	-0.0214 +/- 0.0081
Scatter in the residuals of the lightcurve fit is	0.76 %
Best Comparison Star	#1 - [53, 465]
Optimal Aperture	3.81
Optimal Annulus	9.8
Transit Duration (day)	0.0774 +/- 0.0098

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R fit vs published	0.135 ± 0.012 vs 0.1326 (deviation 0.2σ)
Duration fit vs published	0.0774 d vs 0.0737 d (deviation 0.37σ)
Mid-time O-C	0.7 min
Depth SNR	11.2
Residual scatter	0.76 %

Light curve

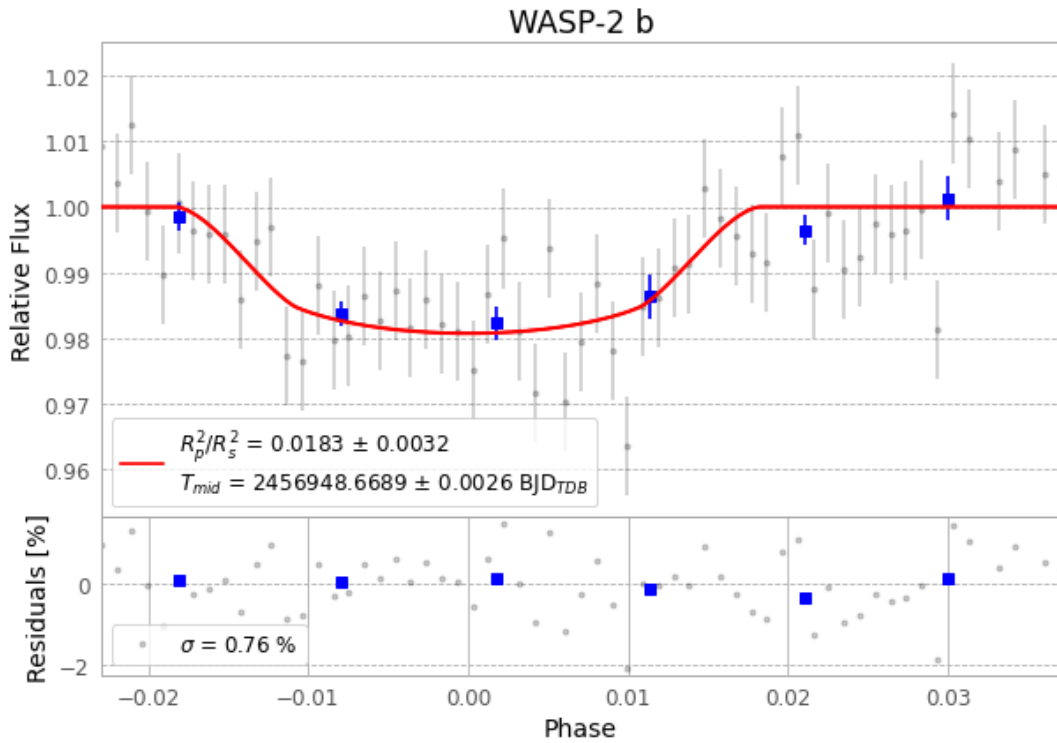


Figure 1 — FinalLightCurve_WASP-2 b_2014-10-17.png (SHA-256 6fc4b2edd19cb655...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [301, 314], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 37dc07ab9afe0964...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

63 FITS frames (41 MB), WASP-2141018024812.FITS → WASP-2141018055412.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-2-141018.log (91a45e822eda...), FinalLightCurve_WASP-2 b_2014-10-17.png (6fc4b2edd19c...), FinalLightCurve_WASP-2 b_2014-10-17.csv (76d03fab58c9...)

Compute resources

Wall time 120.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-30 17:38Z.